Prior Designs:

Ergonomic Improvements in Heavy-Duty Four-Wheel Cart with Pelvis Support"- Advances in Physical Ergonomics and Human Factors Conference paper, SpringerLink Design Attempt 2

Industrial carts require significant pushing/pulling forces causing postural discomfort to shoulder, neck, and lower back operators. This study aimed to diminish postural uneasiness during the operation of the cart by modifying it. The modified cart includes a pelvis support with a waist belt, an increased handle height, and all swivel wheels to scale down the discomfort levels in specific regions and assist with improved cart motion. After modification, the discomfort levels on the shoulder, lower arms, mid back, and lower back are significantly lesser. The results of this study indicate significant differences in discomfort levels and better maneuverability with the modified cart.

(Closest I could find to description)

Utility Cart Mods on LumberJocks Woodworking Forum: Users on this forum have shared their DIY modifications for utility carts, including adjustable casters and flip-up plywood racks

Design Attempt 1

Popular modifications include adding adjustable casters, which allow for height customization and improved stability on uneven surfaces, making the carts more versatile for various tasks. Another

common modification is the installation of flip-up plywood racks, designed to hold large sheets of plywood securely and make transport and storage more convenient.

Design 1